

observed returns and produces true returns that are close to IID (or not forecastable). Note that the variance of the true returns is higher than the observed returns:

$$\text{var}(r_t) = \frac{1 + \phi^2}{1 - \phi^2} \text{var}(r_t^*) \geq \text{var}(r_t^*), \quad (13.3)$$

since $|\phi| < 1$, so we are adding variance to the observed returns to produce estimates of the true returns.

Another way to interpret the unsmoothing process in equations (13.1) and (13.2) is that it is equivalent to assuming that the smoothed, or reported, return follows:

$$r_t^* = (1 - \phi)r_t + \phi r_{t-1}^*, \quad (13.4)$$

and thus the unsmoothed return at time t , r_t^* , is a weighted average of the unsmoothed, or true, return at time t , r_t , and the lagged unsmoothed return in the previous period, r_{t-1}^* . Thus the smoothed returns only slowly update—they partly reflect what is happening in the true returns, but there are lags induced from the appraisal process.

The unsmoothing process has several important properties:

1. Unsmoothing affects only risk estimates and not expected returns.

Intuitively, estimates of the mean require only the first and last price observation (with dividends take “total prices,” which count reinvested dividends).¹² Smoothing spreads the shocks over several periods, but it still counts all the shocks. In Figure 13.2, we can see that the first and last observations are unchanged by infrequent sampling; thus unsmoothing changes only the volatility estimates.

2. Unsmoothing has no effect if the observed returns are uncorrelated.

In many cases, reported illiquid asset returns are autocorrelated because illiquid asset values are appraised. The appraisal process induces smoothing because appraisers use, as they should, both the most recent and comparable sales (which are transactions) together with past appraised values (which are estimated, or perceived, values). The artificial smoothness from the appraisal process has pushed many in real estate to develop pure transactions-based, rather than appraisal-based indexes.¹³ Autocorrelation also results from more

¹²Technically taking means of both the right and left sides in equation (13.2) results in the same means in large samples.

¹³This literature includes both repeat-sales methodologies (see Goetzmann (1992)) and constructing indexes using only transactions (see Gatzlaff and Geltner (1998) and Fisher, Geltner, and Pollakowski (2007)). Some of these methods adjust for the different characteristics of individual homes in creating these indices, like whether an apartment or a house is for sale, whether it is